**Question 1.** Find the vector form or the parametric form for each of the following lines in $\mathbb{R}^3$.

(1) The line passing through $(1, 1, 1)$ and parallel to the vector $(1, 2, 3)$.

Vector form: $\langle 1,1,1\rangle + t\langle 1,2,3\rangle$

Parametric form $\begin{cases} x = 1+t \\ y = 1+2t \\ z = 1+3t \end{cases}$

(2) The line passing through the points $(1, 2, 3)$ and $(4, 5, 6)$.

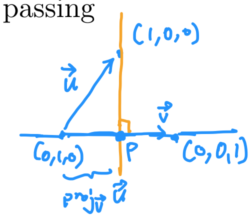
$\langle 4,5,6\rangle - \langle 1,2,3\rangle = \langle 3,3,3\rangle$

Ans: $\langle 1,2,3\rangle + t\langle 3,3,3\rangle$

(3) The intersection of the planes $x + y + z = 1$ and $x - y + z = 1$.

$\begin{cases} x+y+z=1 \\ x-y+z=1 \end{cases} \Leftrightarrow \begin{cases} x+z=1 \\ y=0 \end{cases} \Leftrightarrow \begin{cases} x=t \\ y=0 \\ z=1-t \end{cases}$

(4) The line passing through $(1, 0, 0)$ and intersecting orthogonally with the line passing through $(0, 1, 0)$ and $(0, 0, 1)$.




$\vec{u} = \langle 1,-1,0\rangle$ $\vec{v} = \langle 0,-1,1\rangle$

$proj_{\vec{v}}\vec{u} = \frac{\vec{u}\cdot\vec{v}}{\|\vec{v}\|^2}\vec{v} = \frac{1}{2}\vec{v} = \langle 0,-\frac{1}{2},\frac{1}{2}\rangle$.

$P = \langle 0,1,0\rangle + proj_{\vec{v}}\vec{u} = \langle 0,\frac{1}{2},\frac{1}{2}\rangle$

The line passing through $(1,0,0)$ and $P$ is $\langle 1,0,0\rangle + t\langle -1,\frac{1}{2},\frac{1}{2}\rangle$

(5) The line which is tangent to the curve $\vec{r}(t) = (\cos 2t, \sin 2t, t)$ at $t = \frac{\pi}{4}$.

$\vec{r}(\frac{\pi}{4}) = (\cos\frac{\pi}{2}, \sin\frac{\pi}{2}, \frac{\pi}{4}) = (0,1,\frac{\pi}{4})$

$\frac{d\vec{r}}{dt}(\frac{\pi}{4}) = \langle -2\sin 2t, 2\cos 2t, 1\rangle\Big|_{t=\frac{\pi}{4}} = \langle -2,0,1\rangle$

Ans: $\langle 0,1,\frac{\pi}{4}\rangle + t\langle -2,0,1\rangle$

(6) The line which is tangent to the curve $\vec{r}(t) = (t, t^2, t^3)$ at the point $(1, 1, 1)$.

$\vec{r}(1) = (1,1,1)$

$\frac{d\vec{r}}{dt}(1) = \langle 1, 2t, 3t^2\rangle\Big|_{t=1} = \langle 1,2,3\rangle$

Ans: $\langle 1,1,1\rangle + t\langle 1,2,3\rangle$

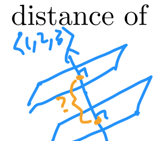
**Question 2.** Match each pair of planes in $\mathbb{R}^3$ with the appropriate relationship between them.

$x + 2y - z = 1$ and $-2x - 4y + 2z = -2$

$x + 2y + 3z = 1$ and $x + 2y + 3z = -2$

$x + 2y + 3z = 1$ and $3x + 2y + 1z = 1$

The planes are parallel with a positive distance of $\frac{3\sqrt{14}}{14}$

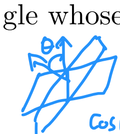



Intersection between $x+2y+3z=1$ with $\langle x,y,z\rangle = t\langle 1,2,3\rangle$ is $(x,y,z) = (\frac{1}{14}, \frac{2}{14}, \frac{3}{14})$

Intersection between $x+2y+3z=-2$ with $\langle x,y,z\rangle = t\langle 1,2,3\rangle$ is $(x,y,z) = (-\frac{2}{14}, -\frac{4}{14}, -\frac{6}{14})$

The distance is $\sqrt{(\frac{3}{14})^2 + (\frac{6}{14})^2 + (\frac{9}{14})^2} = \frac{3}{14}\sqrt{1+2^2+3^2} = \frac{3\sqrt{14}}{14}$

The planes are intersecting with an angle whose cosine is $5/7$




The angle between the planes = the angle between their orthogonal vectors.

$\cos\theta = \frac{\langle 1,2,3\rangle\cdot\langle 3,2,1\rangle}{\|\langle 1,2,3\rangle\|\,\|\langle 3,2,1\rangle\|} = \frac{10}{14} = \frac{5}{7}$

The planes coincide

Points in $x+2y-z=1$ are exactly the points in $-2x-4y+2z=-2$